PROJECT GUIDE · PART 2

AskMyDocs
Technical Documentation

Architecture, backend, data layer (vector DB), API reference, frontend, labs, configuration aur is session ke saare code changes

Solution	`AskMyDocs.slnx` (API + Labs) · Angular app alag
Backend	ASP.NET Core Minimal API, .NET 10 (`net10.0`)
Frontend	Angular 21 (standalone component, signals)
AI layer	Microsoft.Extensions.AI 10.10 · Microsoft Agent Framework 1.23
Vector store	Microsoft.Extensions.VectorData · InMemory / Qdrant
Current AI provider	Ollama (`llama3.2`, `nomic-embed-text`, 768-dim)
Date	15 October 2026

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1. Overview

AskMyDocs ek document Q&A aur IT-support assistant hai. Ye **RAG (Retrieval-Augmented Generation)** aur **tool-calling agent** pattern use karta hai. User text documents upload karta hai. Backend unhe chunks me todta hai, embeddings banata hai aur vector store me save karta hai. Sawaal aane par semantically closest chunks retrieve karke LLM se grounded, cited jawab likhwata hai. Agent mode me LLM khud tools (document search, ticket status, ticket create) chunta hai aur session memory ke saath multi-turn conversation karta hai.

Runnable unit	Type	Path	Entry point	Run
AskMyDocs.Api	ASP.NET Core Web API	`api/AskMyDocs.Api`	`Program.cs`	`dotnet run` → :5080
askmydocs-web	Angular SPA	`web/askmydocs-web`	`src/main.ts`	`npm start` → :4200
AskMyDocs.Labs	.NET console app	`labs/AskMyDocs.Labs`	`Program.cs`	`dotnet run -- lab1..lab7`
Ollama (external)	Local LLM server	installed app	-	auto-start → :11434
Qdrant (optional)	Vector DB container	`docker-compose.yml`	-	`docker compose up -d` → :6333/:6334

Key features

- Document ingestion: paragraph-aware chunking (1200 chars, 200 overlap) + batch embedding + upsert.
- Plain RAG: top-4 semantic retrieval, 0.25 score threshold, numbered citations, "not found" guard.
- Agent: `ChatClientAgent` with 3 tools, per-session memory, tool-usage reporting.
- Streaming: Server-Sent Events (session / tool / token / done events).
- Provider-agnostic: OpenAI, Azure OpenAI ya Ollama, sirf config se.
- Store-agnostic: InMemory ya Qdrant, sirf `VectorStore:Kind` se.

2. Architecture aur request lifecycle

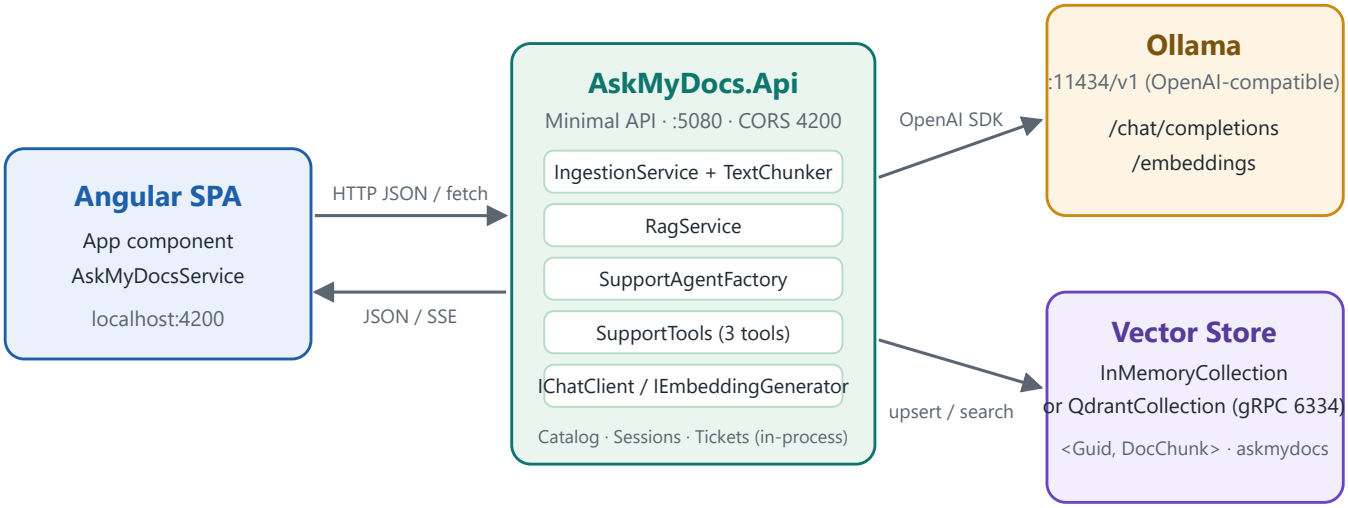


Figure 1: Components, ports aur dependencies

2.1 Layering

Layer	Responsibility	Types
Presentation	UI state, user input, rendering, SSE parsing	`App`, `AskMyDocsService`
HTTP / endpoints	Routing, model binding, SSE writing, CORS	`Program.cs` (Minimal API)
Application services	Ingestion, retrieval, generation, agent orchestration	`IngestionService`, `RagService`, `SupportAgentFactory`, `SupportTools`
AI abstractions	Provider-agnostic chat + embedding	`IChatClient`, `IEmbeddingGenerator<string, Embedding<float>>`
Data abstraction	Provider-agnostic vector CRUD + search	`VectorStoreCollection<Guid, DocChunk>`
Infrastructure	Model hosting, vector persistence	Ollama / OpenAI, InMemory / Qdrant

2.2 Request lifecycle summary

Flow	Sequence
Ingest	UI `file.text()` → `POST /api/documents` → `IngestionService.IngestAsync` → `EnsureCollectionExistsAsync` → `TextChunker.Chunk` → `DocChunk` records → `UpsertAsync` (auto-embedding via generator → Ollama `/embeddings`) → Catalog entry → `IngestResponse`
Plain RAG	`POST /api/ask` → `RagService.AskAsync` → `RetrieveAsync` (`SearchAsync(query, 4)`, query auto-embedded, score ≥ 0.25) → `BuildContext` → `IChatClient.GetResponseAsync` (system + user msg, T=0.1, 800 tokens) → `AskResponse`
Agent	`POST /api/agent/ask` → `CreateAgent` → `GetOrCreateSessionAsync` → `agent.RunAsync` (function-invocation loop: LLM → tool → LLM ...) → collect `FunctionCallContent` names → `AgentAskResponse`

Streaming	`POST /api/agent/stream` → headers `text/event-stream` → `session` event → `RunStreamingAsync` updates → `tool` / `token` events (flush each) → `done`

3. Tech stack aur versions

3.1 Backend NuGet packages

Package	Version	API	Labs	Purpose
`Microsoft.Agents.AI`	1.23.0	✓	✓	Agent framework core (`ChatClientAgent`, `AgentSession`)
`Microsoft.Agents.AI.OpenAI`	1.23.0	✓	✓	OpenAI integration for agents
`Microsoft.Agents.AI.Workflows`	1.23.0		✓	Multi-agent workflows (Lab 7)
`Microsoft.Extensions.AI`	10.10.0	✓	✓	`IChatClient`, `IEmbeddingGenerator`, `AIFunctionFactory`, function invocation
`Microsoft.Extensions.AI.OpenAI`	10.10.1	✓	✓	OpenAI SDK → `IChatClient` adapters (`AsIChatClient`)
`Microsoft.Extensions.VectorData.Abstractions`	10.10.0	✓	(transitive)	`VectorStoreCollection`, attributes
`CommunityToolkit.VectorData.InMemory`	1.0.1	✓	✓	In-memory vector store
`CommunityToolkit.VectorData.Qdrant`	1.0.0	✓		Qdrant connector
`Microsoft.SemanticKernel`	1.81.0		✓	Lab 5 only
`Microsoft.Extensions.Configuration.Json` / `.EnvironmentVariables`	10.0.12		✓	Labs config loading

Note: vector store connectors pehle `Microsoft.SemanticKernel.Connectors.*` naam se aate the, ab `CommunityToolkit.VectorData.*` me hain.

3.2 Frontend (package.json)

Package	Version
`@angular/core`, `common`, `compiler`, `forms`, `platform-browser`, `router`	^21.2.0
`@angular/build`, `@angular/cli`	^21.2.5
`rxjs`	~7.8.0
`typescript`	~5.9.2
`prettier`	^3.8.1

3.3 Runtime environment (is machine par)

Tool	Version / status
.NET SDK	10.0.301 (10.0.201 aur 8.0.419 bhi installed)
Node.js / npm	v24.19.0 / 11.17.0
Ollama	Installed, models: `llama3.2:latest`, `nomic-embed-text:latest`

Docker	Installed nahi (Qdrant ke liye chahiye)
Hardware	19.5 GB RAM, dedicated GPU nahi (CPU inference; Hyper-V display adapter, yaani VM lagti hai)

4. Repository structure

```
AskMyDocs/
├── AskMyDocs.slnx                  Solution: API + Labs (Angular isme nahi)
├── README.md                       Original setup guide (Hinglish)
├── docker-compose.yml              Qdrant service (6333 REST/dashboard, 6334 gRPC)
├── docs/                           Ye documentation (PDF)
├── sample-docs/
│   └── itsm-policy.md              Test document (SLA, escalation, states, access)
├── api/AskMyDocs.Api/
│   ├── Program.cs                  Composition root + endpoints
│   ├── appsettings.json            Ai + VectorStore config
│   ├── appsettings.Development.json  Logging only
│   ├── AskMyDocs.Api.http          Ready-made HTTP requests
│   ├── Properties/launchSettings.json  http :5080, https :7080
│   ├── Models/
│   │   ├── Contracts.cs            Request/response records
│   │   └── DocChunk.cs             Vector record schema
│   ├── Rag/
│   │   ├── TextChunker.cs          Chunking algorithm
│   │   ├── IngestionService.cs     Indexing pipeline + catalog
│   │   └── RagService.cs           Retrieve / augment / generate
│   └── Agents/
│       ├── SupportTools.cs         3 agent tools + demo tickets
│       └── SupportAgentFactory.cs  Agent creation + session store
├── web/askmydocs-web/
│   ├── angular.json, package.json, tsconfig*.json
│   └── src/
│       ├── main.ts, index.html, styles.css
│       └── app/
│           ├── app.ts / app.html / app.css    Root (and only) component
│           ├── app.config.ts                  Providers (HttpClient withFetch)
│           ├── askmydocs.service.ts           API client + SSE reader
│           └── models.ts                      TS contracts
└── labs/AskMyDocs.Labs/
    ├── Program.cs                  lab1..lab7 dispatcher
    ├── AiConfig.cs                 Shared client factory + cosine similarity
    ├── appsettings.json
    └── Lab01_Embeddings.cs ... Lab07_MultiAgent.cs
```

5. Configuration

5.1 `api/AskMyDocs.Api/appsettings.json`

Key	Current value	Code default	Meaning
`Ai:Endpoint`	`http://localhost:11434/v1`	`https://api.openai.com/v1`	OpenAI-compatible base URL
`Ai:ApiKey`	`""`	`not-needed-for-local` (khali/null par)	Provider key. Ollama ko nahi chahiye.
`Ai:ChatModel`	`llama3.2`	`gpt-4o-mini`	Chat/agent model
`Ai:EmbeddingModel`	`nomic-embed-text`	`text-embedding-3-small`	Embedding model (dimension `DocChunk` se match honi chahiye)
`VectorStore:Kind`	`InMemory`	`InMemory`	`InMemory` ya `Qdrant` (case-insensitive)
`VectorStore:Collection`	`askmydocs`	`askmydocs`	Collection name
`VectorStore:QdrantHost`	`localhost`	`localhost`	Qdrant host
`VectorStore:QdrantPort`	`6334`	`6334`	Qdrant gRPC port

5.2 Environment variable override

ASP.NET Core config me environment variables JSON ko override karte hain. Nested key ke liye `__` (double underscore) lagta hai. Labs bhi `AddEnvironmentVariables()` use karte hain.

```
$env:Ai__ApiKey   = "sk-..."          # same terminal me, dotnet run se pehle
$env:Ai__Endpoint = "https://api.openai.com/v1"
$env:VectorStore__Kind = "Qdrant"
```

5.3 Provider switch matrix

Provider	Endpoint	ApiKey	ChatModel	EmbeddingModel	Vector dim
Ollama (current)	`http://localhost:11434/v1`	khali	`llama3.2`	`nomic-embed-text`	**768**
OpenAI	`https://api.openai.com/v1`	`sk-...`	`gpt-4o-mini`	`text-embedding-3-small`	**1536**
Azure OpenAI	`https://<resource>.openai.azure.com/openai/v1`	Azure key	deployment name	deployment name	model ke hisaab se

> **Dimension rule**
> Embedding model badalne par 3 jagah vector dimension badalni padti hai: `Models/DocChunk.cs`, `Lab02_VectorStore.cs` (`Faq`), `Lab04_MiniRag.cs` (`Chunk`). Qdrant use ho raha ho to purana collection delete karna padega, kyunki uski dimension pehli baar banne par fix ho jaati hai.

5.4 Hosting, ports, CORS

Setting	Value	Where
Profile `http`	`http://localhost:5080`, `ASPNETCORE_ENVIRONMENT=Development`	`Properties/launchSettings.json`
Profile `https`	`https://localhost:7080;http://localhost:5080`	`Properties/launchSettings.json`
CORS policy `angular`	Origin `http://localhost:4200`, any header, any method	`Program.cs`
Angular API base URL	`const API = 'http://localhost:5080'` (hard-coded)	`askmydocs.service.ts:8`
Angular dev server	`ng serve` → :4200	`package.json` `start` script
Labs config path	`AppContext.BaseDirectory` (bin me copy, `PreserveNewest`)	`AiConfig.cs`, `.csproj`

Development environment me unhandled exceptions ka poora stack trace 500 response body me aata hai (Developer Exception Page). Debugging ke liye useful hai, production me band hona chahiye.

6. Backend (API) internals

6.1 Startup sequence (`Program.cs`)

1. **Config padho**: `Ai` aur `VectorStore` sections, defaults ke saath.
2. **LLM clients**: ek `OpenAIClient` custom endpoint ke saath. Usse chat aur embedding clients bante hain.
3. **Vector store**: `Kind` ke hisaab se `QdrantCollection` ya `InMemoryCollection`. Dono ko embedding generator diya jata hai.
4. **App services** register karo, aur CORS set karo.
5. **Endpoints** map karo, phir `app.Run()`.

```
var openAi = new OpenAIClient(
    new ApiKeyCredential(apiKey),
    new OpenAIClientOptions { Endpoint = new Uri(endpoint) });

builder.Services.AddSingleton<IChatClient>(_ =>
    openAi.GetChatClient(chatModel)
          .AsIChatClient()
          .AsBuilder()
          .UseFunctionInvocation()   // tool calling ka loop ye chalata hai
          .Build());

builder.Services.AddSingleton<IEmbeddingGenerator<string, Embedding<float>>>(_ =>
    openAi.GetEmbeddingClient(embeddingModel).AsIEmbeddingGenerator());
```

```
builder.Services.AddSingleton<VectorStoreCollection<Guid, DocChunk>>(sp =>
{
    var generator = sp.GetRequiredService<IEmbeddingGenerator<string, Embedding<float>>>();
    if (storeKind.Equals("Qdrant", StringComparison.OrdinalIgnoreCase))
    {
        var client = new QdrantClient(qdrantHost, qdrantPort);
        return new QdrantCollection<Guid, DocChunk>(client, collectionName, ownsClient: true,
            new QdrantCollectionOptions { EmbeddingGenerator = generator });
    }
    return new InMemoryCollection<Guid, DocChunk>(collectionName,
        new InMemoryCollectionOptions { EmbeddingGenerator = generator });
});
```

6.2 Dependency injection

Service	Lifetime	Implementation	Depends on
`IChatClient`	Singleton	OpenAI ChatClient → `AsIChatClient()` → `FunctionInvokingChatClient`	`OpenAIClient`
`IEmbeddingGenerator<string, Embedding<float>>`	Singleton	OpenAI EmbeddingClient → `AsIEmbeddingGenerator()`	`OpenAIClient`
`VectorStoreCollection<Guid, DocChunk>`	Singleton	`InMemoryCollection` / `QdrantCollection`	embedding generator
`IngestionService`	Singleton	class	collection, logger
`RagService`	Singleton	class	collection, `IChatClient`, logger
`SupportTools`	Singleton	class	`RagService`

`SupportAgentFactory`	Singleton	class	`IChatClient`, `SupportTools`

Saari services singleton hain. Isliye catalog, sessions aur in-memory vectors poori app ki life tak (process restart tak) rehte hain.

6.3 Contracts (`Models/Contracts.cs`)

Record	Fields	Used by
`IngestRequest`	`string FileName, string Content`	`POST /api/documents` body
`IngestResponse`	`string DocumentId, string FileName, int ChunkCount`	upload response
`AskRequest`	`string Question, string? SessionId = null`	ask / agent / stream body
`SourceRef`	`int Marker, string FileName, int ChunkIndex, double Score, string Snippet`	citations
`AskResponse`	`string Answer, IReadOnlyList<SourceRef> Sources, int RetrievedCount`	`/api/ask`
`AgentAskResponse`	`string Answer, string SessionId, IReadOnlyList<string> ToolsUsed`	`/api/agent/ask`
`DocumentSummary`	`string DocumentId, string FileName, int ChunkCount, DateTimeOffset UploadedAt`	`GET /api/documents`

JSON camelCase me serialize hota hai (Minimal API web defaults), jaise `fileName`, `chunkCount`. Input case-insensitive hai.

6.4 Chunking algorithm (`Rag/TextChunker.cs`)

Parameter	Value
`DefaultMaxChars`	1200
`DefaultOverlapChars`	200 (must be < maxChars, warna `ArgumentException`)
Min chunk length	30 chars (chhote drop)

```
1. CRLF -> LF; "\n\n" par split -> paragraphs (trimmed, empty removed)
2. har paragraph:
   a. para.Length > maxChars  -> current buffer flush; para ko . ! ? \n par sentences me tod kar
                                 maxChars tak pack karo (har piece alag chunk)
   b. current + para > maxChars (aur current khali nahi)
                              -> current ko chunk bana do; naya current = finished ke aakhri
                                 overlapChars characters + "\n\n"
   c. current += para + "\n\n"
3. last buffer flush
4. trim; length < 30 wale hatao
```

Note: overlap character-based hai, isliye naya chunk kabhi word ke beech se shuru ho sakta hai. Long-paragraph sentence split me original punctuation (`! ?`) `.` se replace ho jaata hai.

6.5 Ingestion (`Rag/IngestionService.cs`)

```
public async Task<IngestResponse> IngestAsync(IngestRequest request, CancellationToken ct = default)
{
    ArgumentException.ThrowIfNullOrWhiteSpace(request.Content);
    await collection.EnsureCollectionExistsAsync(ct);

    var documentId = Guid.NewGuid().ToString("N");
    var pieces = TextChunker.Chunk(request.Content);
    if (pieces.Count == 0) throw new InvalidOperationException("Is document se koi usable chunk nahi
bana.");

    var records = pieces.Select((text, index) => new DocChunk {
        Id = Guid.NewGuid(), DocumentId = documentId, FileName = request.FileName,
        ChunkIndex = index, Text = text, Embedding = text });   // string -> auto-embedded

    await collection.UpsertAsync(records, ct);                 // batch embed + store
    Catalog[documentId] = new DocumentSummary(documentId, request.FileName, pieces.Count,
DateTimeOffset.UtcNow);
    return new IngestResponse(documentId, request.FileName, pieces.Count);
}
```

`DocChunk.Embedding` ka type `string` hai. Collection ke paas `EmbeddingGenerator` hai, isliye `UpsertAsync` ke andar library khud string ko `float[768]` vector me badalti hai. Code me manual "embed then store" step nahi likhna padta.

6.6 Retrieval + generation (`Rag/RagService.cs`)

Constant / option	Value	Effect
`TopK`	4	Max chunks retrieved
`MinScore`	0.25	Is cosine score se neeche wale hits drop (`null` score allowed)
`Temperature`	0.1	Near-deterministic factual output
`MaxOutputTokens`	800	Answer length cap
Snippet	180 chars + "..."	`SourceRef.Snippet`

```
// System prompt (verbatim)
Tum ek document assistant ho. Tumhe neeche CONTEXT diya gaya hai.
Niyam:
1. Jawab SIRF diye gaye CONTEXT se dena hai. Apni general knowledge mat lagao.
2. Har claim ke aage uska source marker lagao, jaise [1] ya [2].
3. Agar CONTEXT me jawab nahi hai to saaf saaf likho:
   "Diye gaye documents me iska jawab nahi mila."
   Andaza bilkul mat lagao.
4. Jawab chhota aur seedha rakho.

// User message format
CONTEXT:
[1] (file: <FileName>, chunk: <ChunkIndex>)
<Text>

[2] ...

SAWAL: <question>
```

```
public async Task<IReadOnlyList<VectorSearchResult<DocChunk>>> RetrieveAsync(string query, int top,
CancellationToken ct = default)
{
    await collection.EnsureCollectionExistsAsync(ct);
    var results = new List<VectorSearchResult<DocChunk>>();
    await foreach (var hit in collection.SearchAsync(query, top, cancellationToken: ct))
        if (hit.Score is null || hit.Score >= MinScore) results.Add(hit);
    return results;
}
```

Zero hits par LLM call skip hoti hai aur fixed message return hota hai: *"Diye gaye documents me iska jawab nahi mila. Pehle koi document upload karo."*

6.7 Agent (`Agents/SupportAgentFactory.cs`, `Agents/SupportTools.cs`)

```
public AIAgent CreateAgent() =>
    new ChatClientAgent(chatClient, instructions: Instructions, name: "AskMyDocsAgent",
        description: "Documents aur ITSM tickets par kaam karne wala support agent",
        tools: tools.AsTools());

public IList<AITool> AsTools() =>
[
    AIFunctionFactory.Create(SearchDocumentsAsync, name: "search_documents"),
    AIFunctionFactory.Create(GetTicketStatus,      name: "get_ticket_status"),
    AIFunctionFactory.Create(CreateTicket,         name: "create_ticket")
];
```

Tool	Parameters	Returns	Backed by
`search_documents`	`query: string`	Top-4 chunks as text: `--- file (chunk n, score x.xxx) ---` + text, ya "Documents me kuch relevant nahi mila."	`RagService.RetrieveAsync`
`get_ticket_status`	`ticketId: string` (e.g. INC-1001)	`Ticket X: status=..., owner=..., priority=...` ya not-found + available IDs	static dictionary
`create_ticket`	`title: string`, `priority: string = "P3"`	`Ticket INC-NNNN ban gaya (priority ...): title`	static dictionary, random id 2000-9998

Demo ticket	Status	Owner	Priority
INC-1001	In Progress	Rahul Verma	P2
INC-1002	Resolved	Priya Nair	P3
INC-1003	Awaiting User	Bhagat Singh	P1

Agent instructions (summary): policy/process/product sawaal par pehle `search_documents` chalao. Ticket ke sawaal par `get_ticket_status`. Naya ticket sirf tab banao jab user saaf bole, aur title/priority confirm karo. Jawab me source file ka naam likho. Jawab na mile to seedha bolo.

Function-invocation loop: LLM response me `FunctionCallContent` aata hai. `FunctionInvokingChatClient` (`UseFunctionInvocation()`) matching `AIFunction` invoke karta hai, result `FunctionResultContent` ke roop me conversation me add karta hai, aur LLM ko dobara call karta hai. Ye tab tak chalta hai jab tak final text response na aa jaye. Parameter schema `AIFunctionFactory` reflection + `[Description]` attributes se banata hai.

Sessions: `ConcurrentDictionary<string, AgentSession> _sessions`. Known `sessionId` par existing session milti hai. Warna `agent.CreateSessionAsync()` se nayi banti hai, aur id = given id ya naya `Guid("N")`. Koi expiry ya size limit nahi hai.

6.8 SSE writer

```
http.Response.Headers.ContentType  = "text/event-stream";
http.Response.Headers.CacheControl = "no-cache";
await WriteEventAsync(http, new { type = "session", sessionId }, ct);
await foreach (var update in agent.RunStreamingAsync(request.Question, session, cancellationToken: ct))
{
    foreach (var call in update.Contents.OfType<FunctionCallContent>())
        await WriteEventAsync(http, new { type = "tool", name = call.Name }, ct);
    if (!string.IsNullOrEmpty(update.Text))
        await WriteEventAsync(http, new { type = "token", text = update.Text }, ct);
}
await WriteEventAsync(http, new { type = "done" }, ct);

static async Task WriteEventAsync(HttpContext http, object payload, CancellationToken ct)
{
    await http.Response.WriteAsync($"data: {JsonSerializer.Serialize(payload)}\n\n", ct);
    await http.Response.Body.FlushAsync(ct);
}
```

7. Data layer: vector DB aur in-memory state

> **Database summary**
> Project me **koi relational database (SQL) nahi hai**. Persistent data ka ek hi candidate hai: **vector store** (InMemory ya Qdrant). Baaki state (catalog, sessions, tickets) process memory me hai.

7.1 Vector record schema: `DocChunk`

Property	C# type	Attribute	Stored as	Purpose
`Id`	`Guid`	`[VectorStoreKey]`	Primary key (Qdrant point id)	Chunk identity
`DocumentId`	`string`	`[VectorStoreData(IsIndexed = true)]`	Payload, indexed	Parent document; filter ke liye
`FileName`	`string`	`[VectorStoreData]`	Payload	Citation display
`ChunkIndex`	`int`	`[VectorStoreData]`	Payload	Document me position
`Text`	`string`	`[VectorStoreData]`	Payload	LLM context + snippet
`Embedding`	`string`	`[VectorStoreVector(768, DistanceFunction = CosineSimilarity)]`	`float[768]` vector	Similarity search

7.2 Store providers

	InMemory (current)	Qdrant
Class	`InMemoryCollection<Guid, DocChunk>`	`QdrantCollection<Guid, DocChunk>`
Location	API process RAM	Docker container `askmydocs-qdrant`, volume `qdrant_data` → `/qdrant/storage`
Connection	-	`QdrantClient(localhost, 6334)` gRPC; REST + dashboard 6333
Persistence	Restart par lost	Restart ke baad bhi rehta hai
Search	Brute-force cosine over all records	ANN index (HNSW)
Enable	`"Kind": "InMemory"`	`docker compose up -d` + `"Kind": "Qdrant"`

7.3 In-process state

State	Type	Owner	Scope	Note
Document catalog	`static ConcurrentDictionary<string, DocumentSummary>`	`IngestionService.Catalog`	Process	Qdrant mode me bhi persist nahi hota. Restart ke baad list khaali hogi, lekin vectors searchable rahenge.
Agent sessions	`ConcurrentDictionary<string, AgentSession>`	`SupportAgentFactory._sessions`	Singleton	No TTL, no cap

Tickets	`static Dictionary<string, (Status, Owner, Priority)>`	`SupportTools.Tickets`	Process	Not thread-safe for concurrent `create_ticket`
Chat turns (UI)	`signal<ChatTurn[]>`	Angular `App`	Browser tab	Refresh par lost

7.4 Example stored record

```
{
  "Id": "7f3c0b1e-...-e21a",
  "DocumentId": "152441e486a9473cbecab2cc5858a81f",
  "FileName": "itsm-policy.md",
  "ChunkIndex": 0,
  "Text": "# WorkElevate ITSM Policy (sample document) ... ## SLA aur Priority ...",
  "Embedding": [0.0213, -0.1134, 0.0871, ...]      // 768 floats (nomic-embed-text)
}
```

8. API reference

Base URL: `http://localhost:5080`. Saare bodies `application/json`. Koi authentication nahi hai.

GET /health

Config check. Kaunsa store aur model chal raha hai.

```
200 OK
{ "status": "ok", "store": "InMemory", "chatModel": "llama3.2", "embeddingModel": "nomic-embed-text" }
```

POST /api/documents

Document ingest karta hai: chunk → embed → store.

```
Request:  { "fileName": "itsm-policy.md", "content": "<poora text>" }
200 OK:   { "documentId": "152441e4...", "fileName": "itsm-policy.md", "chunkCount": 3 }
```

`sample-docs/itsm-policy.md` (~2.7 KB) ke 3 chunks bante hain: 1186, 1178 aur 537 characters. Ye project ka `TextChunker` chalakar verify kiya gaya hai.

Error case	Result
Body invalid JSON / type mismatch	400 (`BadHttpRequestException`)
`content` khali / whitespace	500 (`ArgumentException`, koi validation filter nahi hai)
Koi chunk nahi bana (sab < 30 chars)	500 (`InvalidOperationException`)
Embedding provider error (401, Ollama down, model missing)	500 (e.g. `ClientResultException: HTTP 401`)

GET /api/documents

Indexed documents ki list, newest first.

```
200 OK
[ { "documentId": "152441e4...", "fileName": "formDesign.txt", "chunkCount": 20,
    "uploadedAt": "2026-10-15T09:51:57.092893+00:00" } ]
```

POST /api/ask

Plain RAG: retrieve + generate + citations.

```
Request:  { "question": "P1 ticket ka SLA kitne ghante ka hai?" }
200 OK:   { "answer": "P1 ka resolution SLA 4 ghante hai [1].",
            "sources": [ { "marker": 1, "fileName": "itsm-policy.md", "chunkIndex": 0,
                           "score": 0.7123, "snippet": "# WorkElevate ITSM Policy ..." } ],
            "retrievedCount": 4 }
```

Values illustrative hain. Score aur text model/document par depend karte hain.

POST /api/agent/ask

Agent: tools + session memory, single response.

```
Request:  { "question": "INC-1001 ka status batao aur uski priority ka SLA bhi", "sessionId": null }
200 OK:   { "answer": "...", "sessionId": "9b1f...", "toolsUsed": ["get_ticket_status", "search_documents"]
}

Follow-up (same memory):
Request:  { "question": "aur uska owner kaun hai?", "sessionId": "9b1f..." }
```

POST /api/agent/stream

Agent + Server-Sent Events. Request body `/api/agent/ask` jaisa hi hai. Response `text/event-stream` hai. Har event `data: <json>\n\n` format me aata hai.

Event	Payload	Kab
`session`	`{"type":"session","sessionId":"..."}`	Sabse pehle, ek baar
`tool`	`{"type":"tool","name":"search_documents"}`	Har function call par
`token`	`{"type":"token","text":"..."}`	Har text fragment par
`done`	`{"type":"done"}`	End

Stream shuru hone ke baad error aaye to HTTP status already 200 ho chuka hota hai. Tab stream bina `done` ke toot jaata hai. Koi `error` event type nahi hai.

Testing

`api/AskMyDocs.Api/AskMyDocs.Api.http` me saare requests ready hain. Visual Studio ya VS Code (REST Client extension) me "Send Request" dabao. PowerShell:

```
Invoke-RestMethod http://localhost:5080/health
Invoke-RestMethod http://localhost:5080/api/ask -Method Post -ContentType "application/json" `
  -Body '{"question":"P1 ticket ka SLA kitna hai?"}'
```

9. Frontend (Angular)

9.1 Bootstrapping

```
// main.ts
bootstrapApplication(App, appConfig).catch((err) => console.error(err));

// app.config.ts
providers: [ provideBrowserGlobalErrorListeners(), provideHttpClient(withFetch()) ]
```

Ek hi standalone root component (`App`, selector `app-root`) hai. Router configure nahi hai. `FormsModule` input binding ke liye import hota hai. Template Angular ka naya control flow (`@if`, `@for`) use karta hai.

9.2 Component state (`app.ts`)

Member	Type	Purpose
`documents`	`signal<DocumentSummary[]>`	Sidebar list
`turns`	`signal<ChatTurn[]>`	Chat bubbles
`question`	`signal<string>`	Input box
`mode`	`signal<'rag' | 'agent' | 'stream'>`	Default `'stream'`
`busy`, `error`, `activeTool`	signals	Loading, error banner, "Tool chal raha hai"
`canSend`	`computed`	Non-empty question aur not busy
`totalChunks`	`computed`	Sum of `chunkCount`
`sessionId`	`string | null` (private)	Agent memory id. `clearChat()` par reset hota hai.

Method	Kaam
`refreshDocuments()`	`GET /api/documents`. Fail par "API se connect nahi ho pa raha..."
`onFileSelected(e)`	`file.text()` → `uploadDocument` → refresh → assistant bubble "N chunks bane"
`send()`	User bubble add, phir mode ke hisaab se `sendRag` / `sendAgent` / `sendStream`
`sendStream(q)`	Khaali assistant bubble, phir `for await` events: session → id save, tool → chip + activeTool, token → text append, done → streaming=false
`patch(i, fn)`	Ek turn ko immutably update karta hai (signal ko naya reference chahiye)
`clearChat()`, `setMode()`, `useSample()`	UI helpers

9.3 Template (`app.html`)

- **Sidebar**: file input (`accept=".txt,.md,.json,.csv"`), indexed documents + total chunks, 3 mode buttons, "Chat saaf karo".
- **Chat**: error banner, welcome state + 3 sample questions, turns (tool chips, RAG sources with marker/file/chunk/score/snippet), running-tool indicator, composer form (`ngSubmit` → `send()`).

9.4 Service (`askmydocs.service.ts`)

Method	HTTP	Returns
`uploadDocument(fileName, content)`	`POST /api/documents`	`Promise<IngestResponse>`
`listDocuments()`	`GET /api/documents`	`Promise<DocumentSummary[]>`
`ask(question)`	`POST /api/ask`	`Promise<AskResponse>`
`askAgent(question, sessionId)`	`POST /api/agent/ask`	`Promise<AgentAskResponse>`
`streamAgent(question, sessionId, signal?)`	`POST /api/agent/stream` via `fetch`	`AsyncGenerator<StreamEvent>`

```
// SSE parsing (HttpClient poori response ka wait karta hai, isliye native fetch + ReadableStream)
const reader = response.body.getReader();
const decoder = new TextDecoder();
let buffer = '';
while (true) {
  const { done, value } = await reader.read();
  if (done) break;
  buffer += decoder.decode(value, { stream: true });
  const parts = buffer.split('\n\n');
  buffer = parts.pop() ?? '';                       // adhoora event agle read ke liye
  for (const part of parts) {
    const line = part.trim();
    if (!line.startsWith('data:')) continue;
    yield JSON.parse(line.slice(5).trim()) as StreamEvent;
  }
}
```

9.5 Models (`models.ts`)

`IngestResponse`, `DocumentSummary`, `SourceRef`, `AskResponse`, `AgentAskResponse` C# records ke camelCase mirror hain. UI-only types:

```
export interface ChatTurn { role: 'user' | 'assistant'; text: string;
  sources?: SourceRef[]; toolsUsed?: string[]; streaming?: boolean; }

export type StreamEvent =
  | { type: 'session'; sessionId: string } | { type: 'tool'; name: string }
  | { type: 'token'; text: string }        | { type: 'done' };
```

10. Labs project

`Program.cs` pehle argument (`lab1`..`lab7`) se lab chunta hai. Exception aane par common causes print karta hai (khali key, galat endpoint, dimension mismatch). `AiConfig` `appsettings.json` + env vars padhta hai. Khali key par `"ollama"` placeholder lagata hai. Isme haath se likha cosine similarity function bhi hai.

Lab	Concept	Key APIs	AI?
1 Embeddings	Vector dekho; 4 candidates ko query se rank karo	`GenerateVectorAsync`, `AiConfig.CosineSimilarity`	Embedding
2 Vector store	5 FAQ upsert; semantic search; filter `Category == "Hardware"`	`InMemoryCollection`, `SearchAsync`, `VectorSearchOptions.Filter`	Embedding
3 Chunking	Same doc 200/0, 600/100, 3000/200 par	Local `Chunk()` (simplified copy)	Nahi
4 Mini RAG	Bina RAG vs RAG (top 2, T=0.1)	`GetResponseAsync`, `SearchAsync`	Chat + Embedding
5 Semantic Kernel	Prompt function template; auto function calling	`Kernel.CreateBuilder`, `[KernelFunction]`, `FunctionChoiceBehavior.Auto()`	Chat
6 Agent	Simple agent; tools; session; serialize session	`ChatClientAgent`, `CreateSessionAsync`, `SerializeSessionAsync`	Chat
7 Multi-agent	Triage → Solver → Editor sequential pipeline	`AgentWorkflowBuilder.BuildSequential`, `InProcessExecution.RunStreamingAsync`, `TurnToken`, `WorkflowOutputEvent`	Chat

11. Build, run, test commands

Kaam	Folder	Command
Ollama models (ek baar)	kahin bhi	`ollama pull llama3.2` ; `ollama pull nomic-embed-text`
Ollama check	-	`ollama list` ya browser `http://localhost:11434`
API build	`api/AskMyDocs.Api`	`dotnet build`
API run	`api/AskMyDocs.Api`	`dotnet run` (profile http, :5080)
API run (https)	`api/AskMyDocs.Api`	`dotnet run --launch-profile https`
Solution build	root	`dotnet build AskMyDocs.slnx`
Angular deps	`web/askmydocs-web`	`npm install`
Angular dev	`web/askmydocs-web`	`npm start` (:4200)
Angular prod build	`web/askmydocs-web`	`npm run build` → `dist/`
Angular tests	`web/askmydocs-web`	`npm test` (spec files abhi nahi hain; schematics `skipTests: true`)
Labs	`labs/AskMyDocs.Labs`	`dotnet run -- lab1` … `lab7`
Qdrant	root	`docker compose up -d` / `docker compose down`
Stuck API band karo	-	Ctrl+C, ya `Stop-Process -Name AskMyDocs.Api`

Backend me unit test project nahi hai. Verification ke liye `/health` , `.http` file aur UI flows use hote hain.

12. Change log: is session me kya badla

Kul **7 files** badli gayi hain: 2 bug fixes (Change 1 aur 7), aur 5 changes OpenAI se Ollama par shift karne ke liye. Har change ka purana (BEFORE) aur naya (AFTER) code neeche hai.

Change 1: Khali API key par crash ka fix

File: `api/AskMyDocs.Api/Program.cs` · Line 19 (ab 19-20) · Section "1. Config padho"

Problem: `appsettings.json` me `"ApiKey": ""` tha. `??` operator sirf `null` par fallback deta hai, khali string par nahi. Khali string `ApiKeyCredential` me gayi aur startup par crash hua: `System.ArgumentException: Value cannot be an empty string. (Parameter 'key')`.

BEFORE

```
var apiKey = ai["ApiKey"] ?? "not-needed-for-local";
```

AFTER

```
// Ollama ko key chahiye hi nahi, par ApiKeyCredential khali string nahi leta.
var apiKey = string.IsNullOrWhiteSpace(ai["ApiKey"]) ? "not-needed-for-local" : ai["ApiKey"]!;
```

Labs ka `AiConfig.CreateOpenAIClient()` pehle se `IsNullOrWhiteSpace` check karta tha. Ab API bhi wahi karti hai.

Change 2: API ko Ollama par point kiya

File: `api/AskMyDocs.Api/appsettings.json` · Lines 10-15 (`"Ai"` section)

Wajah: OpenAI key available nahi thi. Upload par `HTTP 401 invalid_api_key` aa raha tha. Ollama free aur local hai. GPU na hone ki wajah se 8B `llama3.1` ki jagah 3B `llama3.2` chuna, jo tool calling bhi support karta hai.

BEFORE

```
"Ai": {
  "Endpoint": "https://api.openai.com/v1",
  "ApiKey": "",
  "ChatModel": "gpt-4o-mini",
  "EmbeddingModel": "text-embedding-3-small"
},
```

AFTER

```
"Ai": {
  "Endpoint": "http://localhost:11434/v1",
  "ApiKey": "",
  "ChatModel": "llama3.2",
  "EmbeddingModel": "nomic-embed-text"
},
```

Change 3: Labs ko Ollama par point kiya

File: `labs/AskMyDocs.Labs/appsettings.json` · Lines 2-8 (`"Ai"` section, `_comment` unchanged)

BEFORE

```
"Ai": {
  "_comment": "OpenAI ke liye apni key daalo. Ollama ke liye Endpoint badlo aur ApiKey khali chhod do.",
  "Endpoint": "https://api.openai.com/v1",
  "ApiKey": "",
  "ChatModel": "gpt-4o-mini",
  "EmbeddingModel": "text-embedding-3-small"
}
```

AFTER

```
"Ai": {
  "_comment": "OpenAI ke liye apni key daalo. Ollama ke liye Endpoint badlo aur ApiKey khali chhod do.",
  "Endpoint": "http://localhost:11434/v1",
  "ApiKey": "",
  "ChatModel": "llama3.2",
  "EmbeddingModel": "nomic-embed-text"
}
```

Change 4: DocChunk vector dimension 1536 → 768

File: `api/AskMyDocs.Api/Models/DocChunk.cs` · Lines 38-44 (`Embedding` property aur uska comment)

Wajah: `nomic-embed-text` 768-dimension vector banata hai. Schema 1536 rehta to "dimension mismatch" error aata.

BEFORE

```
/// <summary>
/// 1536 = text-embedding-3-small ki dimension.
/// Agar tum Ollama ka nomic-embed-text use karoge to ye 768 karna padega,
/// warna store "dimension mismatch" error dega.
/// </summary>
[VectorStoreVector(1536, DistanceFunction = DistanceFunction.CosineSimilarity)]
public string Embedding { get; set; } = string.Empty;
```

AFTER

```
/// <summary>
/// 768 = Ollama ke nomic-embed-text ki dimension.
/// Agar OpenAI ka text-embedding-3-small use karoge to ye 1536 karna padega,
/// warna store "dimension mismatch" error dega.
/// </summary>
[VectorStoreVector(768, DistanceFunction = DistanceFunction.CosineSimilarity)]
public string Embedding { get; set; } = string.Empty;
```

Change 5: Lab 2 (Faq record) dimension 1536 → 768

File: `labs/AskMyDocs.Labs/Lab02_VectorStore.cs` · Line 37 (class `Faq`)

BEFORE

```
// Is string ka embedding banega aur isi par search hogi.
[VectorStoreVector(1536, DistanceFunction = DistanceFunction.CosineSimilarity)]
public string Embedding { get; set; } = "";
```

AFTER

```
// Is string ka embedding banega aur isi par search hogi.
[VectorStoreVector(768, DistanceFunction = DistanceFunction.CosineSimilarity)]
public string Embedding { get; set; } = "";
```

Change 6: Lab 4 (Chunk record) dimension 1536 → 768

File: `labs/AskMyDocs.Labs/Lab04_MiniRag.cs` · Line 26 (class `Chunk`)

BEFORE

```
[VectorStoreVector(1536, DistanceFunction = DistanceFunction.CosineSimilarity)]
public string Embedding { get; set; } = "";
```

AFTER

```
[VectorStoreVector(768, DistanceFunction = DistanceFunction.CosineSimilarity)]
public string Embedding { get; set; } = "";
```

Change 7: Lab 7 multi-agent workflow chal hi nahi raha tha

File: `labs/AskMyDocs.Labs/Lab07_MultiAgent.cs` · Lines 82-101 (workflow run aur event loop)

Problem: `dotnet run -- lab7` 3 second me khatam ho jata tha aur kisi agent ka output nahi aata tha. Diagnosis me do wajah mili:

1. Workflow ke agents messages jama karte hain aur `TurnToken` **milne par hi LLM call karte hain**. Code token nahi bhejta tha, isliye pehla agent bina LLM call ke complete hua aur workflow ruk gaya.
2. Token bhejne ke baad bhi `AgentResponseEvent` emit nahi hota. Agents `AgentResponseUpdateEvent` (streaming tukde) bhejte hain, aur poori conversation end me `WorkflowOutputEvent` me aati hai.

BEFORE

```
var run = await InProcessExecution.RunStreamingAsync(workflow, input);

await foreach (var evt in run.WatchStreamAsync())
{
    // Har agent apna kaam khatam karte hi ek AgentResponseEvent deta hai.
    if (evt is AgentResponseEvent done)
    {
        Console.WriteLine($"\n### {done.Response.AgentId ?? "agent"} ka output ###");
        Console.WriteLine(done.Response.Text);
        Console.WriteLine(new string('-', 72));
    }
}
```

AFTER

```
var run = await InProcessExecution.RunStreamingAsync(workflow, input);

// Workflow ke agents messages jama karte rehte hain aur TurnToken milne par hi
// LLM call karte hain. Ye token na bhejo to workflow bina kuch kiye khatam ho jata hai.
await run.TrySendMessageAsync(new TurnToken(emitEvents: true));

await foreach (var evt in run.WatchStreamAsync())
{
    // Workflow khatam hone par poori conversation ek WorkflowOutputEvent me aati hai.
    // Har agent ka message AuthorName (agent ka naam) ke saath hota hai.
    if (evt is WorkflowOutputEvent output && output.Data is IEnumerable<ChatMessage> messages)
    {
        foreach (var message in messages.Where(m => m.Role == ChatRole.Assistant))
        {
            Console.WriteLine($"\n### {message.AuthorName ?? "agent"} ka output ###");
            Console.WriteLine(message.Text);
            Console.WriteLine(new string('-', 72));
        }
    }
}
```

Verify kiya: fix ke baad TriageAgent, SolverAgent aur EditorAgent teeno ka output print hua (Ollama `llama3.2`, CPU par ~126 sec). Output end me ek saath aata hai, live streaming nahi hoti.

Code ke alawa kiye gaye kaam

Action	Detail
Build verification	API aur Labs dono shuru me bina error build hue. Labs Ollama changes ke baad bhi build hua (0 warnings, 0 errors).
Startup verification	Change 1 ke baad API start hui aur `/health` ne `200 OK` diya.
Upload diagnosis	Sample document `POST /api/documents` par bheja. Response ke stack trace me `HTTP 401 invalid_api_key` mila.
Locked process stop	Purani API (PID 23948, `AskMyDocs.Api.exe`) band ki, jo build ko `MSB3027` se rok rahi thi.
Documentation	`docs/` folder banaya, ye 2 PDF yahan hain.

Wapas OpenAI par jaana ho to

1. Dono `appsettings.json` me Endpoint `https://api.openai.com/v1`, ChatModel `gpt-4o-mini`, EmbeddingModel `text-embedding-3-small` karo. Key `$env:Ai__ApiKey` se do.
2. `DocChunk.cs`, `Lab02_VectorStore.cs`, `Lab04_MiniRag.cs` me 768 ko 1536 karo.
3. API restart karo. InMemory store khaali shuru hoga, isliye documents dobara upload karo.

13. Known limitations aur production recommendations

Limitation	Impact	Recommendation
Saara state in-memory	Restart par documents, sessions, naye tickets lost	Qdrant (ya Azure AI Search / pgvector) + SQL catalog + Redis sessions
Catalog Qdrant mode me bhi persist nahi hota	Restart ke baad list khaali, lekin purane vectors search me aate rahenge	Catalog SQL me rakho, ya Qdrant payload se derive karo
Authentication / authorization nahi	Network par koi bhi upload/query kar sakta hai	JWT / Entra ID, per-tenant `DocumentId` /tenant filter
Delete / re-index endpoint nahi	Same file dobara upload par duplicate chunks	`DELETE /api/documents/{id}` (`DocumentId` indexed hai, filter se delete); content hash se dedupe
Sirf plain text files	PDF/DOCX upload nahi hote (browser `file.text()`)	Server-side multipart upload + extraction (PdfPig, OpenXML)
Validation errors 500 dete hain	Client ko galat status code milta hai	Input validation + `ProblemDetails` (400)
Hard-coded URLs / CORS	Deploy karne par code badalna padega	Angular environment files ya dev proxy; CORS origins config se
Sessions unbounded	Memory growth; lambi history par token cost aur latency badhti hai	TTL, max turns, summarization; Redis
`create_ticket` par hard guard nahi	LLM bina confirmation ke ticket bana sakta hai (sirf prompt-level rule hai)	Human-in-the-loop approval; real ITSM API idempotency ke saath
Demo tickets: static non-thread-safe dictionary, random id	Concurrency issues, id collision possible	Real repository / ITSM integration
Stream error handling	Mid-stream failure par UI ko error event nahi milta	`error` SSE event + UI handling; `AbortController` se cancel
`MinScore = 0.25` OpenAI embeddings ke liye socha gaya tha	`nomic-embed-text` ka score distribution alag ho sakta hai	Apne documents par recall/precision dekhkar retune karo
CPU inference, 3B model	Slow responses; tool selection kam reliable. Example: Lab 5 me model ne `GetTicketStatus` ka result dekhe bina `GetSlaHours(P1)` call kiya, jabki INC-1001 P2 hai, aur galat SLA (4 ghante) bataya. Lab 7 me Triage ne "exactly 2 line" format follow nahi kiya.	GPU machine, bada model, ya hosted provider
Docs drift	README abhi bhi OpenAI ko default batata hai; Lab 1 console text me "1536 numbers" likha hai (actual dimension upar print hoti hai)	README / text update karo

14. Troubleshooting

Symptom	Root cause	Fix
Startup: `Value cannot be an empty string. (Parameter 'key')`	Khali `ApiKey`, sirf null check	Fixed (Change 1)
Upload 500, body me `HTTP 401 (invalid_api_key)`	OpenAI endpoint, key nahi	Key do, ya Ollama config (Change 2)
500 + connection refused `:11434`	Ollama nahi chal raha	Ollama app start karo / `ollama serve`
500 + model not found	Model pull nahi hua	`ollama pull llama3.2`, `ollama pull nomic-embed-text`
Dimension mismatch	Schema dim ≠ model dim	Section 5.3 matrix follow karo
`MSB3027` / `MSB3021` file locked	Purani API instance chal rahi hai	Us terminal me Ctrl+C, ya `Stop-Process -Name AskMyDocs.Api`
`Couldn't find a project to run`	Galat folder	API folder me `dotnet run`, Angular me `npm start`
UI: "API se connect nahi ho pa raha"	API down, ya CORS/port mismatch	API :5080 par chalao; UI sirf :4200 se allowed hai
`/health` purana model dikhata hai	Config change ke baad restart nahi hua	API restart karo
Jawab "documents me nahi mila" jabki text hai	Score < 0.25, ya chunk me context toot gaya	`MinScore` / chunk size tune karo; `RetrieveAsync` output debug karo